

Shakya Jayakody, Ph.D.

Curriculum Vitae

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RESEARCH INTERESTS

- **Hardware-Software Co-Design for Deep Learning:** Optimized inference serving architectures, fault-tolerant Large Language Model (LLM) serving, and high-availability distributed systems.
- **High-Performance GPU Computing:** Acceleration of custom CUDA kernels, optimized memory architectures, and sparse matrix multiplication (SpGMM) on NVIDIA GPU architectures (A100, H100, H200).
- **Out-of-Core Computing for GCNs:** Row block-wise alignment partitioning, compressed data structures, and dynamic double-buffered PCIe data transfers overlapping compute and communication.
- **Computer Architecture & Memory Systems:** Memory-bounded architectural improvements, CSR-CSC matrix multiplication acceleration, and hybrid DRAM/Non-Volatile Memory (NVM) rank-based page replacement systems.

EDUCATION

University of Central Florida Orlando, FL
Ph.D. in Electrical Engineering Aug. 2020 – Dec. 2025
Dissertation: *“High-Performance Computational Kernels for Algorithm-System Co-Design in Machine Learning: Enabling Efficient GCN Training and LLM Inference Serving”*
Advisor: Dr. Jun Wang, FIEEE

University of Central Florida Orlando, FL
M.S. in Computer Engineering Jan. 2024 – Dec. 2024
Specialization: Machine Learning and Computer Architecture
Advisor: Dr. Mingjie Lin

Louisiana Tech University Ruston, LA
M.S. in Electrical Engineering Dec. 2018 – May 2020
Specialization: Robotics, Automation Control Systems, and Microsystems
Advisor: Dr. Chester G. Wilson

Louisiana Tech University Ruston, LA
B.S. in Electrical Engineering (Minor in Mathematics) Sep. 2013 – May 2016

ACADEMIC APPOINTMENTS & EXPERIENCE

Postdoctoral Researcher Dec. 2025 – May 2026
Computer Systems Architecture and Data Science Lab, University of Central Florida Orlando, FL

- **Distributed Diffusion Transformer (DiT) Scalability:** Formulated distributed training and inference scaling strategies for DiT-AR, a text-to-video generation framework. Designed multi-GPU pipeline schedules to enhance temporal coherence and optimize multi-node high-performance cluster computing throughput.
- **Academic Proposal Development:** Prepared grant proposal documentation targeting high-performance machine learning hardware, GPU architectures, and algorithmic systems co-design, ensuring compliance with federal agency guidelines.

Graduate Research Assistant Aug. 2020 – Dec. 2025
Computer Systems Architecture and Data Science Lab, University of Central Florida Orlando, FL

- **High-Performance Sparse GPU Kernels:** Developed custom CUDA computational kernels accelerating Sparse General Matrix-Matrix Multiplication (SpGMM), bypassing GPU memory bottlenecks through architectural and algorithmic co-design.
- **Out-of-Core GCN Training Platforms:** Created the AIRES and Peridot frameworks. Devised row block-wise alignment partitioning of compressed matrices and custom CUDA tiling algorithms to overlap host-to-device data transfer over PCIe, achieving a 2.5x speedup in resource-constrained CPU-GPU GCN training.

- **Fault-Tolerant LLM Inference Systems:** Co-developed GhostServe, a shadow-checkpointing serving system for distributed GPU nodes. Realized a 2.7x backup speedup and 2.1x recovery speedup, sustaining continuous high availability and low-latency client request serving during multi-node GPU crashes.
- **High-Performance Supercomputing Benchmarks:** Benchmarked large-scale GPU scaling limits and memory-access latency on NVIDIA H100, H200, and A100 nodes at the NSF-funded Delta AI Lab (NCSA).

GRANTS & FUNDING PROPOSAL DEVELOPMENT

Approved Cyberinfrastructure Grants & Supercomputing Allocations

- **NSF ACCESS Supercomputing Allocation CIS250473** (Approved, 2025 – 2026)
Principal Proposal Writer (PI: Dr. Jun Wang, UCF)
 - Successfully drafted and secured a major supercomputer resource allocation from the NSF Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program.
 - Obtained high-performance GPU partition allocations on the **Delta** and **DeltaAI** GPU supercomputers (NCSA) to support large-scale distributed AI serving (GhostServe) and out-of-core GCN training (AIRES) benchmarks.
 - Aligned allocation metrics directly with research milestones for **NSF Award #2400014** (*SHF: Small: Taming Huge Page Problems for Memory Bulk Operations Using a Hardware/Software Co-Design Approach*).
- **PSC Neocortex Program Wafer-Scale Compute Proposal** (Approved, 2026) – *Lead Proposal Writer*
 - Prepared and submitted a research access proposal to the Pittsburgh Supercomputing Center (PSC) Neocortex Program to evaluate GhostServe’s shadow-checkpointing serving system on Cerebras CS-2 wafer-scale engine platforms.

Pending Grants & Cluster Resource Proposals

- **AMD University Program Equipment/Access Proposal** (Pending, 2026) – *Lead Proposal Writer*
 - Submitted a proposal for high-performance cluster access on AMD MI300X nodes to port the GhostServe runtime to the AMD ROCm open-source software platform.
 - Formulated plans for custom HIP kernel implementations for XOR/RDP/Reed-Solomon encoding and Infinity Fabric/RCCL communication.

Proposal Co-Authoring & General Development

- **Proposal Co-Authoring & Technical Drafting:** Actively assisted in conceptualizing, drafting, and developing competitive research proposals submitted to federal funding agencies (including the National Science Foundation - NSF - CCF/CNS divisions) and corporate research sponsors.
- **Technical Architecture & Roadmap Formulation:** Designed multi-GPU system topology diagrams, formulated hardware-contained scheduling milestones, and generated preliminary experimental scaling results on H100/A100 clusters.
- **Institutional Compliance:** Coordinated with the Office of Research at the University of Central Florida to ensure proposals aligned with strict federal compliance, budgetary allocations, and institutional capabilities.

PUBLICATIONS

Refereed Journal Articles Under Review / In Preparation

- [J1] [TPDS] Jayakody, S., Arain, H., Nehate, C., Zhao, Y., & Wang, J. Rachnar: Fault-Tolerant Multi-Node Disaggregated LLM Serving. *IEEE Transactions on Parallel and Distributed Systems* (TPDS). (Under Review, 2026).

Refereed Conference Publications

- [C1] [MLSys'26] Jayakody, S.*, Zhao, Y.*, Nehate, C., & Wang, J. GhostServe: A Lightweight Checkpointing System in the Shadow for Fault-Tolerant LLM Serving. *Proceedings of the 2026 Conference on Machine Learning and Systems* (MLSys). (* Equal contribution.) (Accepted)
- [C2] [ICMLA'25] Pinnock, A.*, Jayakody, S.*, Roxy, K. A., & Ahmed, Md. R. EdgeProfiler: A Fast Profiling Framework for Lightweight LLMs on Edge Using Analytical Model. *Proceedings of the 2025 IEEE 24th International Conference on Machine Learning and Applications* (ICMLA), pp. 814-819, doi: 10.1109/ICMLA66185.2025.00119.

(* Equal contribution.)

- [C3] [ASAP'25] Jayakody, S., Zhao, Y., & Wang, J. AIRES: Accelerating Out-of-Core GCNs via Algorithm-System Co-Design. *Proceedings of the 2025 IEEE 36th International Conference on Application-specific Systems, Architectures and Processors (ASAP)*, pp. 1-8, doi: 10.1109/ASAP65064.2025.00011.
- [C4] [CIC'23] Jayakody, S., & Wang, J. EMBARK: Memory Bounded Architectural Improvement in CSR-CSC Sparse Matrix Multiplication. *Proceedings of the 2023 IEEE 9th International Conference on Collaboration and Internet Computing (CIC)*, pp. 8-17, doi: 10.1109/CIC58953.2023.00012.

Refereed Workshop & Symposia Papers

- [W1] [MLArchSys'24] Jayakody, S., & Wang, J. Peridot: Accelerating Out-of-Core GCN Data Reuse Pattern and Co-Design on GPU. In *Machine Learning for Computer Architecture and Systems Workshop (MLArchSys)* at the IEEE International Symposium on Computer Architecture (ISCA 2024).

Refereed Preprints

- [U1] [ISCA] Zhao, Y., Jayakody, S., Price, D., Wu, D., & Wang, J. Compute More, Save More: Reliable and Sustainable LLM Inference. *IEEE International Symposium on Computer Architecture (ISCA)* Preprint (Under Review, 2026).

TEACHING & PEDAGOGICAL EXPERIENCE

Lead Course Instructor (Lab Sections)

Department of Electrical and Computer Engineering, University of Central Florida

Orlando, FL

- **EEL 4742C: Embedded Systems Lab** (Spring 2022, Fall 2022, Spring 2023, Summer 2023, Spring 2024)
Instructed laboratory sessions focused on microcontrollers, hardware interfacing, and real-time operating systems. Designed experiment structures, graded projects, and mentored undergraduate designs.
- **EEL 3701C: Digital Systems Lab** (Fall 2020, Summer 2021)
Taught digital system fundamentals, logic design, Verilog prototyping, and FPGA synthesis.
- **EEL 3307C: Electronics I Lab** (Fall 2021)
Supervised experimental lab sessions on analog circuits, operational amplifiers, and semiconductor devices.
- **EEL 3801C: Computer Organization Lab** (Fall 2023, Spring 2024, Summer 2024)
Delivered instructional sessions on assembly language, ALU design, memory hierarchy, and pipeline architecture.
- **EEL 4914C / 4915C: Senior Electrical and Computer Engineering Design Lab** (Summer 2024, Fall 2024, Spring 2025)
Mentored senior design teams on complex system integration, PCB schematic design, power systems, and final prototyping milestones.

Graduate Teaching Assistant

Department of Electrical and Computer Engineering, University of Central Florida

Orlando, FL

- Led exam preparation reviews, graded coursework, and hosted weekly consultation office hours for both core Electrical Engineering and Computer Engineering/Machine Learning tracks:
 - **Topics in Machine Learning** (Fall 2024)
 - **Introduction to Deep Learning** (Spring 2025)
 - **Computer Architecture** (Fall 2021)
 - **Massive Storage and Big Data** (Spring 2022, Fall 2022, Fall 2024)
 - **Digital Signal Processing Fundamentals** (Summer 2023)
 - **Engineering Analysis and Computation** (Spring 2021, Summer 2021, Fall 2022)
 - **Computer Organization** (Fall 2023)
 - **Engineering Applications of Intelligent Systems** (Spring 2025)

Invited Guest Lectureships

University of Central Florida

Orlando, FL

- Delivered specialized research-driven guest lectures for advanced undergraduate and graduate level courses:
 - *Data Intensive Computing* (EEL 6760) – Topic: GPU Architecture & Sparse Accelerations Spring 2026
 - *Massive Storage and Big Data* (EEL 4798) – Topic: High Availability in LLM Cluster Serving Spring 2026
 - *Computer Architecture* (EEL 4768) – Topic: Out-of-Core Compute and PCIe Communication Summer 2026

STUDENT ADVISING & MENTORING RECORD

University of Central Florida

Orlando, FL

- **Ashvik Vijai** (Undergraduate Student) Jan. 2025 – Sep. 2025
Mentored on custom CUDA programming, profiling sparse computations on GPUs, and out-of-core sparse data representations.

HONORS & AWARDS

- **Outstanding Analytical Skills Award**, Graduate School Dean, Louisiana Tech University 2020
- **Top Student Ambassador Award**, International Student Director, Louisiana Tech University 2020
- **Endowed Superior Graduate Student Scholarship**, Louisiana Tech University 2019 – 2020
- **Bulldog Out-of-State Scholarship**, Louisiana Tech University 2013 – 2016

PROFESSIONAL EXPERIENCE (INDUSTRY)

Technical Director

Lanka Telephone Co. (Pvt) Ltd.

Dec. 2017 – Nov. 2018

Colombo, Sri Lanka

- Managed corporate engineering budgets, technical bids, and tender procurement cycles for enterprise communication client installations. Streamlined equipment supply workflows, accelerating project completion times by 15%.

Lead Electrical & Telecommunication Engineer

Lanka Telephone Co. (Pvt) Ltd.

May 2017 – Dec. 2017

Colombo, Sri Lanka

- Led multi-disciplinary teams in installing enterprise telecom networks. Created and deployed comprehensive safety compliance procedures in line with municipal regulations.

Electrical & Telecommunication Engineer

Lanka Telephone Co. (Pvt) Ltd.

Jan. 2017 – May 2017

Colombo, Sri Lanka

- Supervised engineering teams on multi-million dollar installation projects. Prepared capital resource and labor forecasts presented to the Board of Directors for approval.

Software Engineer

Lanka Telephone Co. (Pvt) Ltd.

July 2016 – Jan. 2017

Colombo, Sri Lanka

- Developed commercial telephone billing systems in C/C++ and corporate stock tracking tools in C#.NET.

TECHNICAL SKILLS & RESEARCH INFRASTRUCTURE

- **Architectural Simulators & Software:** Gem5, Ramulator, Vivado HLx, Xilinx IDE, PyTorch, PyCUDA, Verilog.
- **High-Performance GPU Clusters:** Large-scale model training and profiling on NVIDIA H100, H200, and A100 GPU clusters via the NSF-funded Delta AI Lab.
- **Research Assets:** RTX 5090 GPU primary workstation (deep learning benchmarking), RTX 3070 GPU secondary workstation, and Digilent Nexys 3 FPGA prototyping boards.
- **Programming Languages:** CUDA, C/C++, Python, MIPS Assembly.
- **Circuit & 3D Design:** Schema and PCB design with Eagle and Fusion 360.

PROFESSIONAL CERTIFICATIONS & TRAINING

- Information Security Awareness Training, University of Central Florida 2025

- Responsible Conduct of Research for Engineers, CITI Program 2020
- Research Misconduct & Integrity Training, University of Central Florida 2020